

Strategic coexistence theory for evolutionary games

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Abstract

Evolutionary game theory and ecological coexistence theory both seek to predict the outcome of competition between biological entities, be they strategies or species, but the two fields have relied on largely separate approaches. Replicator equations provide a foundation for analyzing strategy competition and the stability of strategic equilibria, yet they do not explicitly separate the mechanisms that stabilize competition from those that equalize fitness differences between strategies. Here, we extend modern coexistence theory from community ecology to develop strategic coexistence theory (SCT), a framework for quantifying strategic niche and fitness differences between competing strategies. SCT recovers the classic classification of two-strategy games, distinguishing competitive exclusion, coexistence, and priority effects within a shared niche–fitness difference space. Applying SCT to five mechanisms for the evolution of cooperation further reveals that these mechanisms promote cooperation through distinct dynamical routes: kin selection, network reciprocity, and group selection primarily reduce fitness differences, whereas direct and indirect reciprocity destabilize competition and generate priority effects. Finally, applying SCT to microbial public-goods game shows that nonlinear microbial growth can both stabilize and equalize competition between cooperators and defectors, allowing coexistence. Together, these results show that SCT provides a complementary framework for comparing evolutionary games and teasing apart the coexistence mechanisms underlying strategy competition.

Introduction

Understanding the mechanisms that promote the evolutionary stability of competing strategies is a fundamental aim in evolutionary biology (Lewontin, 1961; Slobodkin, 1964; Smith and Price, 1973; Taylor and Jonker, 1978). To this end, evolutionary game theory (EGT) and replicator dynamics have provided key foundations, allowing us to predict the outcome of strategic competition through payoff inequalities, invasion criteria, and stability analysis (Hofbauer and Sigmund, 1998; Cressman and Tao, 2014). However, less attention has been given to similarities between replicator dynamics and community ecology until recently (Tarnita and Traulsen, 2025;

Allesina, 2026), even though the classic replicator equation stems from the generalized Lotka-Volterra equation (Page and Nowak, 2002).

Interestingly, even a simple two-strategy game can yield heterogeneous outcomes that mirror outcomes in community ecology (Hauert, 2002; Szabó and Fath, 2007; Cooney, 2020; Tarnita and Traulsen, 2025): the dominance of a single strategy (prisoner’s dilemma and harmony games), coexistence (hawk–dove games), and bistability (stag-hunt games). Such differences in conditions that promote evolutionary stability and dominance of a single strategy and those that permit strategy coexistence indicate that EGT can be understood from the perspective of coexistence theory (Motro, 1991; Doebeli and Hauert, 2005; Gore et al., 2009). Recasting game dynamics as a coexistence problem immediately raises the following question: when two strategies coexist, does the corresponding mechanism stabilize or equalize competition? Even though EGT allows us to predict whether two strategies can coexist or not, a framework for quantifying coexistence mechanisms between competing strategies remains elusive.

Modern coexistence theory (MCT) from community ecology presents a particularly useful framework for quantifying coexistence mechanisms (Chesson, 2000; Godoy et al., 2014; Godoy and Levine, 2014; Kraft et al., 2015). MCT predicts that coexistence of two competing entities, be they species or strategies, requires stabilizing niche differences to overcome average fitness differences. In community ecology, stabilizing niche differences represent differences in factors that limit the growth of competing species, causing each species to limit itself more than it limits its competitor (Chesson, 2000). In EGT, a strategy’s niche can be interpreted as strategic environments defined by the composition of competing strategies and the resulting interactions. Thus, stabilizing niche differences arise when each strategy is favored by a different strategic environment, such that the spread of each strategy creates conditions that limit its own relative growth and favor the recovery of its competitor. In contrast, fitness differences represent differences in the ability of each entity to compete under the limiting environments created by its competitor (Chesson, 2000). This differs from payoff differences in EGT, which depend on the underlying strategy frequency and therefore combine both stabilizing effects caused by changes in the strategic environment and residual fitness differences that favor one strategy over another.

So far, MCT has been primarily utilized for understanding species coexistence, but recent work showed that MCT can be extended beyond species competition, especially for studying pathogen strain competition, allowing one to directly quantify niche and fitness differences of competing entities, be they species or strains (Sieben et al., 2022; Park et al., 2024, 2026). Inspired by this effort, we present a Strategic Coexistence Theory (SCT) for predicting coexistence and exclusion of competing strategies in EGT by quantifying strategic niche and fitness differences. We begin by introducing SCT, which can be applied to any generic system of EGT described by replicator equations. We show that SCT can recover outcomes from classic, two-strategy games, providing reinterpretation of classic results from coexistence-

theoretic perspective. We then apply SCT to comparing five mechanisms for the evolution of cooperation, revealing that direct and indirect reciprocity destabilize competition between cooperation and defection, thus pushing the system towards priority effect regime. Finally, we consider a case study of microbial cooperation, where nonlinear microbial growth can both stabilize and equalize competition between cooperation and defection, allowing coexistence of strategies.

Strategic coexistence theory

Here, we present a brief derivation of strategic coexistence theory (SCT) (Figure 1). To do so, we begin with a classic replicator equation, which allows us to model changes in the frequency of competing strategies based on their payoffs (Hofbauer and Sigmund, 1998; Cressman and Tao, 2014):

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (1)$$

where x_i represents the frequency of strategy i , $\pi_i(\mathbf{x})$ represents the payoff of strategy i for a given strategy distribution $\mathbf{x}$, and $\bar{\pi}(\mathbf{x}) = \sum_i \pi_i(\mathbf{x})x_i$ represents the average payoff. Since strategy frequencies must sum to 1 (i.e., $\sum_i x_i = 1$), a single-strategy equilibrium dominated by strategy i can be described by a unit vector $\mathbf{e}_i$, where all elements are equal to zero except for the i -th element. We use these single-strategy equilibria to define niche and fitness differences between two competing strategies i and j .

Because payoffs in many games can be negative, payoff ratios can be difficult to interpret. Thus, we work on a positive multiplicative scale by exponentiating the payoff terms:

$$W_i(\mathbf{x}) = \exp(\pi_i(\mathbf{x})). \quad (2)$$

This exponential transformation is convenient because it preserves the payoff ordering and does not alter the invasion conditions. Therefore, this allows us to compare strategic environments through ratios, as in modern coexistence theory. For convenience, we refer to $W_i(\mathbf{x})$ as the multiplicative payoff of strategy i .

First, we define the niche overlap ρ between strategies i and j as the ratio between the geometric mean of each strategy's multiplicative payoff in the strategic environment that it creates when common, $\sqrt{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}$, and the geometric mean of each strategy's multiplicative payoff in the environment generated by its competitor, $\sqrt{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}$. The corresponding niche difference is then given by

$$1 - \rho = 1 - \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \quad (3)$$

When $1 - \rho > 0$, each strategy performs better, on average, in the environment generated by its competitor than in the environment generated by itself. In other

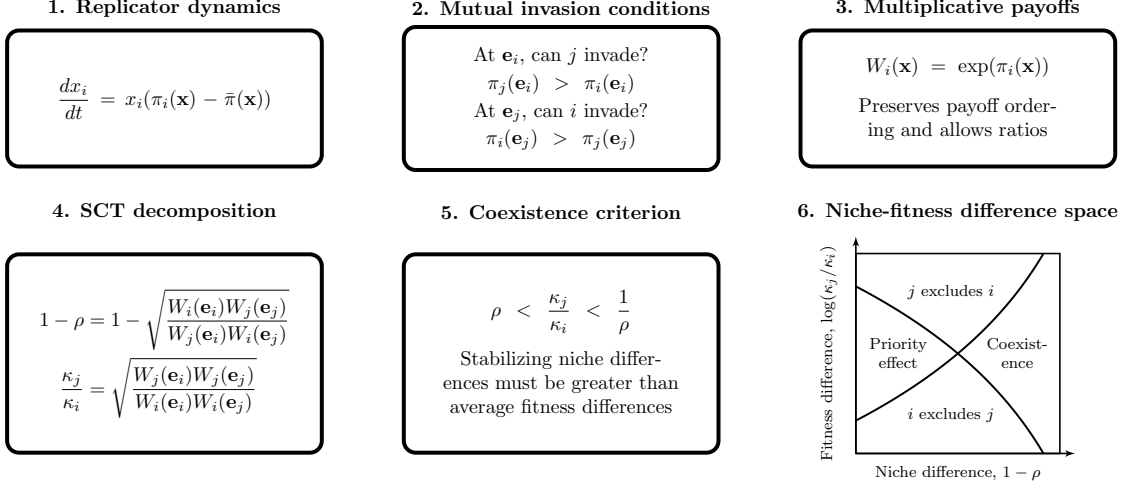


Figure 1: **A schematic diagram summarizing strategic coexistence theory.** Strategic coexistence theory decomposes pairwise replicator dynamics into stabilizing and equalizing mechanisms. Mutual invasion conditions can be rewritten on a multiplicative payoff scale, allowing us to derive strategic niche difference $1 - \rho$ and fitness difference κ_j/κ_i . Then, the inequality $\rho < \kappa_j/\kappa_i < 1/\rho$ determines mutual invasion.

words, intraspecific competition is stronger than interspecific competition. Second, we define the fitness difference as the ratio between the geometric means of the multiplicative payoff of each strategy across the two single-strategy environments:

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \quad (4)$$

In other words, strategy j has an average competitive advantage over strategy i ($\kappa_j/\kappa_i > 1$) when it performs better, on average, across the environments generated by itself and its competitor.

Together, for any two strategies described by the classic replicator equation, mutual invasion requires stabilizing niche differences to overcome average fitness differences. Specifically, the condition for mutual invasion can be rewritten as (Chesson, 2000):

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (5)$$

This inequality applies to any games described by the classic replicator equation because ρ and κ_j/κ_i can be calculated directly from ratios of multiplicative payoffs. This decomposition not only allows us to understand how different mechanisms contribute to the coexistence or exclusion of competing strategies but also provides shared axes for comparing different games.

127 We note that the absolute magnitudes of niche and fitness differences are sensitive
 128 to payoff scaling because they are defined on an exponentiated payoff scale. However,
 129 the qualitative outcome of strategic competition under SCT is preserved because the
 130 exponential transformation does not alter invasion conditions.

131 Classic two-strategy games

132 As an illustrative example, we begin with classic two-strategy games (Hauert, 2002;
 133 Szabó and Fath, 2007; Cooney, 2020; Tarnita and Traulsen, 2025), which can be
 134 described by the following payoff matrix (Figure 2A):

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix}. \quad (6)$$

135 Here, the first and second rows represent strategies 1 and 2, respectively, and the
 136 columns represent the strategy of the interacting opponent. Writing x_1 to denote
 137 the frequency of strategy 1, the expected payoffs are

$$\pi_1(\mathbf{x}) = Rx_1 + S(1 - x_1), \quad (7)$$

$$\pi_2(\mathbf{x}) = Tx_1 + P(1 - x_1). \quad (8)$$

138 Thus, the dynamics can be written as

$$\frac{dx_1}{dt} = x_1(1 - x_1) [(R - T)x_1 + (S - P)(1 - x_1)]. \quad (9)$$

139 The signs of $T - R$ and $S - P$ determine whether each strategy can invade the
 140 other when rare. Specifically, strategy 2 can invade strategy 1 when $T > R$, whereas
 141 strategy 1 can invade strategy 2 when $S > P$. These two invasion conditions partition
 142 the game space into four regimes (Figure 2B): prisoner's dilemma ($T > R > P >$
 143 S), hawk-dove game ($T > R, S > P$), stag-hunt game ($R > T, P > S$), and
 144 harmony game ($R > T, S > P$). In standard EGT, these outcomes are obtained by
 145 analyzing the stability of equilibrium conditions. For simple games, this analysis is
 146 straightforward.

147 A coexistence-theoretic approach recovers the same classification but provides
 148 a different interpretation (Figure 2C). For a two-strategy game determined by this
 149 payoff matrix, the niche difference and fitness difference are given by

$$1 - \rho = 1 - \exp\left(\frac{R + P - S - T}{2}\right), \quad (10)$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{T + P - R - S}{2}\right). \quad (11)$$

150 As shown above, mutual invasion requires $\rho < \kappa_2/\kappa_1 < 1/\rho$, which is equivalent to
 151 $T > R$ and $S > P$. This corresponds to the hawk-dove game, in which each strategy

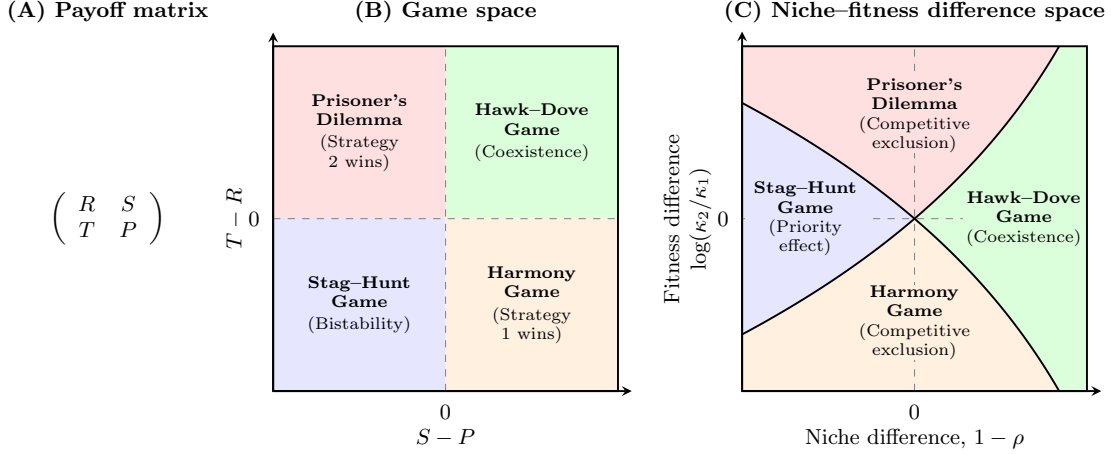


Figure 2: **Coexistence-theoretic interpretation of classic two-strategy games.** (A) General payoff matrix for a two-strategy game, where rows represent the focal strategy and columns represent the strategy of the interacting opponent. (B) Classification of four classic games based on the payoff differences $T - R$ and $S - P$. Prisoner’s dilemma and harmony games lead to the dominance of one strategy; hawk–dove games allow coexistence; and stag–hunt games result in bistability. (C) Mapping of the same four games onto niche–fitness difference space. Hawk–dove games correspond to the coexistence regime, where stabilizing niche differences overcome fitness differences. Prisoner’s dilemma and harmony games correspond to competitive-exclusion regimes, whereas stag–hunt games correspond to the priority-effect regime. $R > P$ is assumed throughout, but this does not affect coexistence outcome in any way.

152 can invade when rare and the two strategies coexist. The remaining games also
 153 have natural interpretations in niche–fitness difference space. Prisoner’s dilemma
 154 and harmony games represent competitive exclusion regimes, where one strategy
 155 has a fitness advantage that exceeds niche differences. In contrast, stag–hunt games
 156 represent a priority-effect regime, where neither strategy can invade the other and
 157 the outcome depends on initial conditions. Therefore, coexistence theory not only
 158 allows us to recover the classic results from two-strategy games but also provides new
 159 interpretations for these results in terms of niche and fitness differences. Building on
 160 this, we next apply this framework to comparing core mechanisms of cooperation.

161 Comparisons of mechanisms for the evolution of co- 162 operation

163 Even though cooperative behavior can be found across a wide range of biological
 164 systems, a simple prisoner’s dilemma shows that cooperation is not an evolution-

arily stable strategy (May, 1987). Thus, many researchers have sought to identify mechanisms that promote the evolution of cooperation (West et al., 2007). Here, we compare five classic mechanisms for the evolution of cooperation using SCT and ask how they differ from the perspective of strategy coexistence.

Specifically, we consider the following mechanisms: kin selection, direct reciprocity, indirect reciprocity, network reciprocity, and group selection (Figure 3A). Kin selection favors cooperation when two interacting individuals are genetic relatives, where the genetic relatedness r must be greater than the cost-to-benefit ratio: $r > c/b$ (Hamilton, 1964). Direct reciprocity favors cooperation when current cooperation may promote future cooperation by the same individual through repeated encounters (Trivers, 1971; Axelrod and Hamilton, 1981). In this case, the probability of repeated interaction (i.e., “next round”) must be greater than the cost-to-benefit ratio: $w > c/b$. Indirect reciprocity favors cooperation when current cooperation may promote future cooperation by different individuals through reputation (Nowak and Sigmund, 1998; Wedekind and Milinski, 2000; Nowak and Sigmund, 2005). Thus, the social acquaintanceship (i.e., the probability of knowing someone’s reputation) must be greater than the cost-to-benefit ratio: $q > c/b$. Network reciprocity favors cooperation via clustering of cooperators, requiring the benefit-to-cost ratio to be greater than the average number of neighbors: $b/c > k$ (Nowak and May, 1992; Lieberman et al., 2005). Finally, group selection favors cooperation via multilevel selection, where groups with more cooperators exhibit faster growth at the group level even if defectors reproduce faster at the individual level (Traulsen and Nowak, 2006). This requires $b/c > 1 + (n/m)$, where n is the maximum group size and m is the number of groups. All of these mechanisms can be described by a simple 2×2 payoff matrix (Nowak, 2006), from which the corresponding niche and fitness differences can be computed using Eq. (10) and Eq. (11).

First, the prisoner’s dilemma provides a baseline scenario, where defection is evolutionarily stable (Figure 3A). From the perspective of SCT, there is complete niche overlap, and therefore defection competitively excludes cooperation (Figure 3B). Building on this, SCT shows that five mechanisms promote evolutionary stability of cooperation via different dynamical routes (Figure 3B). For example, kin selection allows the fitness of cooperation to increase as genetic relatedness r increases (Figure 3B). However, it does not affect the niche difference between cooperation and defectors. Similarly, network reciprocity and group selection modulate the fitness of cooperation without causing any changes in strategic niches. In contrast, direct and indirect reciprocity destabilize competition while increasing the fitness of cooperation, thereby pushing the competition outcome towards a priority effect regime. Equivalently, these two mechanisms promote bistability of defection and cooperation.

SCT further reveals another difference between direct and indirect reciprocity. In particular, direct reciprocity is unable to generate sufficient fitness difference to competitively exclude defection even when the probability of next round w reaches 1. In contrast, indirect reciprocity is able to generate sufficient fitness difference to push the system towards the boundary between priority-effect regime and competitive-

(A)

Prisoner's dilemma	Kin selection
$\begin{pmatrix} b-c & -c \\ b & 0 \end{pmatrix}$	$\begin{pmatrix} (b-c)(1+r) & br-c \\ b-r & 0 \end{pmatrix}$
Direct reciprocity	Indirect reciprocity
$\begin{pmatrix} (b-c)/(1-w) & -c \\ b & 0 \end{pmatrix}$	$\begin{pmatrix} b-c & -c(1-q) \\ b(1-q) & 0 \end{pmatrix}$
Network reciprocity	Group selection
$\begin{pmatrix} b-c & H-c \\ b-H & 0 \end{pmatrix}$	$\begin{pmatrix} (b-c)(m+n) & (b-c)m-cn \\ bn & 0 \end{pmatrix}$

(B)

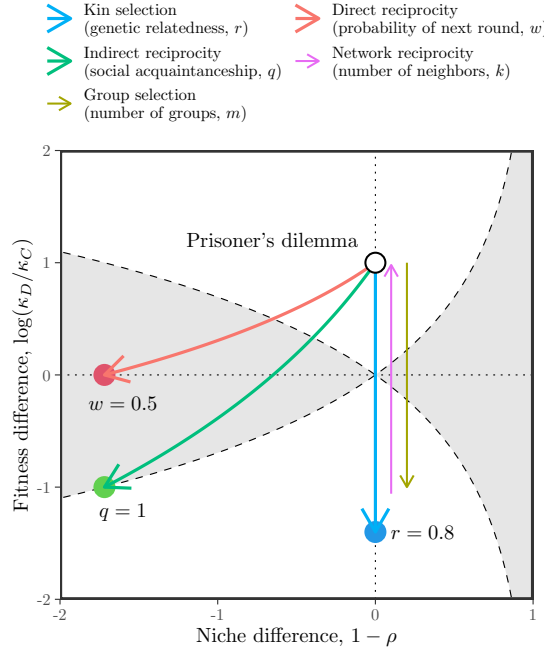


Figure 3: **Comparisons of five mechanisms for the evolution of cooperation using SCT.** (A) Payoff matrices for the baseline scenario (prisoner's dilemma) and five mechanisms for the evolution of cooperation (Nowak, 2006). Prisoner's dilemma is characterized by the benefit for the recipient b and the cost for the donor c . Network reciprocity is parameterized based on $H = [(b-c)k - 2c]/[(k+1)(k-2)]$. (B) Changes in niche and fitness differences driven by each mechanism. Each arrow represents the direction of change as we increase the corresponding parameter, indicated in figure legends. Note that kin selection, network reciprocity, and group selection all result in changes in fitness difference without causing any changes in niche difference. Therefore, we indicate the direction of change for network reciprocity and group selection separately to avoid overlapping arrows.

exclusion regime as social acquaintanceship q reaches 1, indicating that an additional mechanism will be able to promote complete dominance of cooperation.

We note that none of these mechanisms allow coexistence of cooperation and defection. Therefore, we move on to the final example, applying SCT to microbial cooperation, where nonlinear benefits and environmental context can promote coexistence of cooperators and defectors.

Microbial cooperation and public goods

Here, we consider a simple, phenomenological model of competition between cooperating and cheating yeast in a cooperative sucrose-metabolism system, where invertase

217 production by cooperators allows cheaters to utilize glucose, a public good released
 218 through sucrose hydrolysis (Carlson and Botstein, 1982; Dickinson, 1998; Gore et al.,
 219 2009). Specifically, invertase production by cooperators incurs a cost c and gener-
 220 ates a total benefit of 1 that is captured privately with efficiency ϵ . Assuming a
 221 linear relationship between glucose availability and microbial growth, the payoffs for
 222 cooperators π_C and cheaters (defectors) π_D can then be written as

$$\pi_C(f) = \epsilon + f(1 - \epsilon) - c, \quad (12)$$

$$\pi_D(f) = f(1 - \epsilon), \quad (13)$$

223 where the use of glucose depends on the fraction of cooperators f . The dynamics of
 224 cooperators and defectors are described by the following replicator equation:

$$\frac{df}{dt} = f(1 - f) [\pi_C(f) - \pi_D(f)] \quad (14)$$

$$= f(1 - f)(\epsilon - c) \quad (15)$$

225 For this linear growth model, the payoff difference is fixed to $\epsilon - c$ independent of
 226 f (Figure 4A). Therefore, cooperation is favored when efficiency is greater than the
 227 cost of cooperation, $\epsilon > c$, whereas defection is favored when the cost is greater than
 228 efficiency, $c > \epsilon$ (Gore et al., 2009).

229 Our framework allows us to reinterpret this result from the perspective of coexis-
 230 tence theory. In particular, the two strategies have no niche difference, corresponding
 231 to complete niche overlap, $\rho = 1$. In contrast, the fitness difference is determined by
 232 the balance between private efficiency and the cost of cooperation:

$$\frac{\kappa_D}{\kappa_C} = \exp(c - \epsilon). \quad (16)$$

233 Therefore, this model always predicts competitive exclusion, except in the neutral
 234 case $c = \epsilon$. The outcome is determined entirely by the fitness difference between
 235 cooperators and defectors, rather than by stabilizing niche differences. This example
 236 resembles the prisoner's dilemma discussed in the previous example (Figure 3).

237 In real biological systems, the effect of glucose on microbial growth can follow a
 238 nonlinear relationship:

$$\pi_C(f) = [\epsilon + f(1 - \epsilon)]^\alpha - c, \quad (17)$$

$$\pi_D(f) = [f(1 - \epsilon)]^\alpha, \quad (18)$$

239 where $\alpha = 0.15$ reflects the experimental relationship between glucose and microbial
 240 growth (Gore et al., 2009). Then, the dynamics of cooperators and defectors are
 241 described by the following replicator equation:

$$\frac{df}{dt} = f(1 - f) [\{\epsilon + f(1 - \epsilon)\}^\alpha - c - \{f(1 - \epsilon)\}^\alpha]. \quad (19)$$

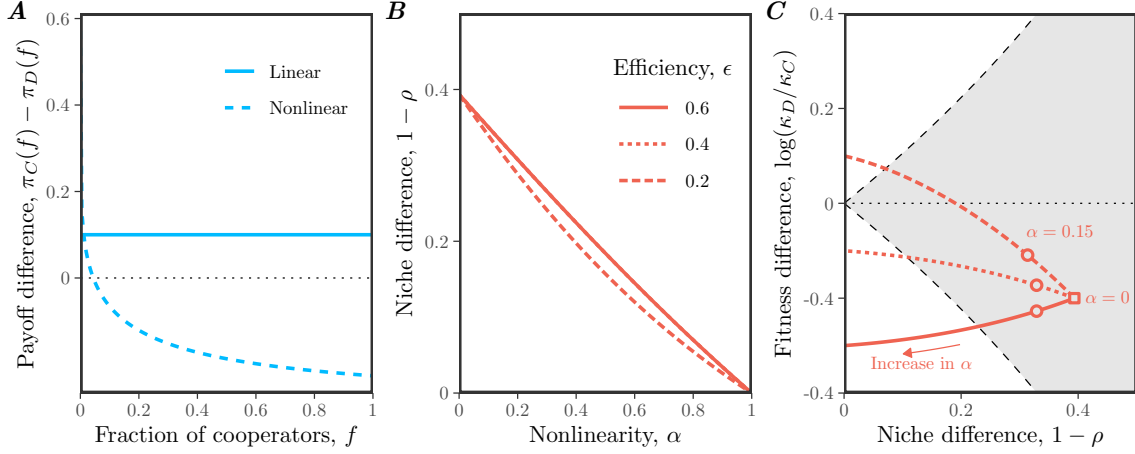


Figure 4: **Effects of nonlinear microbial growth on coexistence of cooperation and defection.** (A) Payoff differences for linear and nonlinear models of microbial growth. In panel A, we assumed $\epsilon = 0.4$, $c = 0.3$, and $\alpha = 0.15$ for illustrative purposes. (B) Effect of nonlinearity α on niche difference $1 - \rho$. Each line assumes a different value of efficiency ϵ . Note that lines for $\epsilon = 0.6$ and $\epsilon = 0.4$ are completely overlapping. (C) Effect of nonlinearity α on both niche difference $1 - \rho$ and fitness difference $\log(\kappa_D/\kappa_C)$. The gray shaded area represents the coexistence regime. The square represents the condition when $\alpha \rightarrow 0^+$. The circles represent the condition when $\alpha = 0.15$, which correspond to the experimental relationship between glucose and microbial growth (Gore et al., 2009). In panels B and C, we assumed $c = 0.3$, while allowing ϵ and α to vary.

Under this nonlinear assumption, the payoff difference decreases with the cooperator fraction f , which can promote the coexistence of cooperators and defectors (Figure 4A): cooperators have a competitive advantage when rare, whereas defectors have a competitive advantage when cooperators are abundant.

From a coexistence-theoretic perspective, this means that nonlinear growth can generate sufficient stabilizing niche difference to overcome the fitness difference. In particular, the niche and fitness differences for this model correspond to:

$$1 - \rho = 1 - \exp\left(\frac{1 - \epsilon^\alpha - (1 - \epsilon)^\alpha}{2}\right), \quad (20)$$

$$\frac{\kappa_D}{\kappa_C} = \exp\left(c + \frac{(1 - \epsilon)^\alpha - 1 - \epsilon^\alpha}{2}\right). \quad (21)$$

These expressions clarify the separate roles of cost, efficiency, and nonlinearity. For example, the cost of cooperation does not affect niche difference and only affects fitness difference. Instead, niche difference is generated by the concave relationship between glucose availability and microbial growth, which increases from 0 to $1 - \exp(-1/2)$ as α goes from 1 to 0 (Figure 4B,C). In contrast, as $\alpha \rightarrow 0^+$, the fitness

254 difference approaches

$$\log(\kappa_D/\kappa_C) \rightarrow c - \frac{1}{2}, \quad (22)$$

255 which no longer depends on the private capture efficiency ϵ (Figure 4C). Thus, strong
256 concavity not only generates stabilizing niche differences, but also acts as an equal-
257 izing mechanism (Figure 4C).

258 Discussion

259 Standard evolutionary game theory and replicator equations provide powerful tools
260 for predicting the outcome of strategy competition, including strategic dominance,
261 coexistence, and bistability (Hofbauer and Sigmund, 1998; Cressman and Tao, 2014).
262 However, these outcomes are typically understood by analyzing payoff differences
263 within individual games, which makes it difficult to compare coexistence mechanisms
264 across models. Building on Chesson’s Modern Coexistence Theory from community
265 ecology (Chesson, 2000), we developed a strategic coexistence theory (SCT) that
266 allows us to decompose strategy competition in terms of stabilizing niche differences
267 and fitness differences. This decomposition provides a unifying framework for com-
268 paring different games from the perspective of coexistence theory, allowing us to
269 identify stabilizing and equalizing mechanisms for strategy competition.

270 To demonstrate the utility of SCT, we considered three case studies. First, the
271 analysis of classic two-strategy games showed that SCT recovers the original clas-
272 sification for these classic games, highlighting the similarities between evolutionary
273 games and community ecology (Hauert, 2002; Szabó and Fath, 2007; Cooney, 2020;
274 Tarnita and Traulsen, 2025). Then, we compared different mechanisms for the evo-
275 lution of cooperation, which showed that each mechanism promotes cooperation via
276 different dynamical routes (Nowak, 2006). Kin selection, network reciprocity, and
277 group selection primarily affect the fitness difference between cooperation and defec-
278 tion, whereas direct and indirect reciprocity can destabilize competition and generate
279 priority effects. Finally, our analysis of microbial cooperation showed that coexis-
280 tence between cooperators and defectors requires more than reducing the fitness cost
281 of cooperation: nonlinear benefits can generate stabilizing niche differences while
282 also reducing fitness asymmetry (Gore et al., 2009). Taken together, these examples
283 show that SCT can provide new insights into ecological underpinnings of strategy
284 competition.

285 There are several limitations to SCT. First, similar to MCT from community
286 ecology, SCT only allows for comparisons of pairwise strategies and therefore cannot
287 determine the outcomes of games with more than two strategies (Chesson, 2000).
288 Second, we focused on classic replicator equations, but Tarnita and Traulsen (2025)
289 recently pointed out that these classic formulations implicitly ignore differences in
290 intrinsic growth rates. In such cases, replicator equations may fail to predict the out-
291 come of competition (Tarnita and Traulsen, 2025), and a full Lotka-Volterra model is
292 needed to describe the abundances, rather than frequencies, of strategies. Thus, SCT

293 inherits the same assumptions as the classic replicator framework: it is most appro-
294 priate when strategy competition can be described in terms of relative frequencies
295 alone. When strategies differ intrinsically in growth, MCT may be required instead
296 to tease apart mechanisms of strategy coexistence. Finally, we have focused on simple
297 examples to illustrate the utility of SCT, but many games involve additional com-
298 plexities, including population structure (Nowak et al., 2010), stochasticity (Wang
299 et al., 2025; Bodin et al., 2026), and explicit ecological feedback. Extending SCT to
300 these settings will be necessary to determine how broadly stabilizing and equalizing
301 mechanisms can be compared across evolutionary games.

302 In conclusion, SCT provides a bridge between evolutionary game theory and
303 community ecology, allowing strategy competition to be understood from the per-
304 spective of coexistence theory. Therefore, rather than replacing existing EGT or
305 replicator-dynamics frameworks, SCT provides a complementary tool for teasing
306 apart the coexistence mechanisms underlying evolutionary games. More broadly,
307 our study builds on recent efforts to extend MCT beyond species competition (Park
308 et al., 2024, 2026), providing a foundation for building unifying coexistence theory
309 for competing entities, including species, pathogen strains, and behavioral strategies.

310 Supplementary Text

311 S1 Derivation of the strategic coexistence theory

312 To derive strategic coexistence theory (SCT), we begin with replicator equations:

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (\text{S1})$$

313 where x_i represents the frequency of strategy i , π_i represents the payoff of strategy
 314 i , and $\bar{\pi}$ represents the average payoff. Then, the mutual invasion condition for two
 315 competing strategies i and j can be written as

$$\pi_j(\mathbf{e}_j) < \pi_i(\mathbf{e}_j), \quad (\text{S2})$$

$$\pi_i(\mathbf{e}_i) < \pi_j(\mathbf{e}_i). \quad (\text{S3})$$

316 These inequalities will be preserved when we exponentiate the payoffs, $W_i(\mathbf{x}) =$
 317 $\exp(\pi_i(\mathbf{x}))$:

$$W_j(\mathbf{e}_j) < W_i(\mathbf{e}_j), \quad (\text{S4})$$

$$W_i(\mathbf{e}_i) < W_j(\mathbf{e}_i). \quad (\text{S5})$$

318 Rearranging, we obtain the following inequality:

$$\frac{W_i(\mathbf{e}_i)}{W_j(\mathbf{e}_i)} < 1 < \frac{W_i(\mathbf{e}_j)}{W_j(\mathbf{e}_j)}. \quad (\text{S6})$$

319 By multiplying $\sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}$ on both sides, we obtain:

$$\sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}} < \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}} < \sqrt{\frac{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}} \quad (\text{S7})$$

320 Thus, by defining niche overlap ρ and fitness difference κ_j/κ_i as follows,

$$\rho = \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}, \quad (\text{S8})$$

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}, \quad (\text{S9})$$

321 we obtain the following inequality that defines conditions for mutual invasion and
 322 long-term coexistence of two competing strategies:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (\text{S10})$$

323 S2 Applications to classic two-strategy games

324 Consider a two-strategy game described by the following payoff matrix:

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix}. \quad (\text{S11})$$

325 There are four classic games that we can define from this simple payoff matrix:
 326 prisoner's dilemma ($T > R > P > S$), hawk-dove game ($T > R, S > P$), stag-hunt
 327 game ($R > T, P > S$), and harmony game ($R > T, S > P$). Here, we show that
 328 each game corresponds to a distinct region in the niche-fitness difference space under
 329 SCT.

330 To do so, we first begin by deriving expressions for niche overlap ρ and fitness
 331 difference κ_j/κ_i . Writing x_1 to denote the frequency of strategy 1, the expected
 332 payoff for each strategy corresponds to:

$$\pi_1(\mathbf{x}) = Rx_1 + S(1 - x_1), \quad (\text{S12})$$

$$\pi_2(\mathbf{x}) = Tx_1 + P(1 - x_1). \quad (\text{S13})$$

333 Then, multiplicative payoffs for each strategy when rare can be written as:

$$W_1(1) = \exp(R) \quad (\text{S14})$$

$$W_1(0) = \exp(S) \quad (\text{S15})$$

$$W_2(1) = \exp(T) \quad (\text{S16})$$

$$W_2(0) = \exp(P) \quad (\text{S17})$$

334 Therefore, we have:

$$\rho = \exp\left(\frac{R + P - S - T}{2}\right), \quad (\text{S18})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{T + P - R - S}{2}\right). \quad (\text{S19})$$

335 As before, SCT allows us to predict the mutual invasion of two strategy using the
 336 following inequality:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (\text{S20})$$

337 Then, simple algebra reveals that each condition in mutual invasion inequality
 338 can be equivalently described by the payoff inequality:

$$\rho < \frac{\kappa_2}{\kappa_1} \iff R < T \quad (\text{S21})$$

$$\rho > \frac{\kappa_2}{\kappa_1} \iff R > T \quad (\text{S22})$$

$$\frac{\kappa_2}{\kappa_1} < \frac{1}{\rho} \iff P < S \quad (\text{S23})$$

$$\frac{\kappa_2}{\kappa_1} > \frac{1}{\rho} \iff P > S \quad (\text{S24})$$

339 These inequalities then allow us to classify outcomes of two-strategy games from the
 340 perspective of SCT. Note that the inequality between R and P does not affect the
 341 outcome.

342 **S3 Applications to mechanisms for the evolution of coopera-** 343 **tion**

344 We consider five mechanisms for the evolution of cooperation, all of which can be
 345 represented as a simple 2×2 payoff matrix (Nowak, 2006). Here, we go through each
 346 mechanism and derive measures for niche and fitness differences.

347 **Prisoner's dilemma.** Here, we consider a special case of Prisoner's dilemma
 348 which can be described by the following payoff matrix:

$$\begin{pmatrix} b - c & -c \\ b & 0 \end{pmatrix}, \quad (\text{S25})$$

349 where b represents the benefit for the recipient and c represents cost for the donor.
 350 In this case, the niche and fitness differences correspond to:

$$1 - \rho = 0 \quad (\text{S26})$$

$$\frac{\kappa_2}{\kappa_1} = \exp(c). \quad (\text{S27})$$

351 Therefore, as long as the cost is greater than zero, prisoner's dilemma always results
 352 in the competitive exclusion of cooperation.

353 **Kin selection.** Kin selection can be described by the following payoff matrix:

$$\begin{pmatrix} (b - c)(1 + r) & br - c \\ b - rc & 0 \end{pmatrix}, \quad (\text{S28})$$

354 where r represents the genetic relatedness. In this case, the niche and fitness differ-
 355 ences correspond to:

$$1 - \rho = 0 \quad (\text{S29})$$

$$\frac{\kappa_2}{\kappa_1} = \exp(c - br). \quad (\text{S30})$$

356 Therefore, defection will exclude cooperation when $c > br$. Equivalently, cooperation
 357 will be favored when $r > c/b$, which recovers the classic result.

358 **Direct reciprocity.** Direct reciprocity can be described by the following payoff
 359 matrix:

$$\begin{pmatrix} (b - c)/(1 - w) & -c \\ b & 0 \end{pmatrix}, \quad (\text{S31})$$

where w represents the probability of next round. In this case, the niche and fitness differences correspond to:

$$1 - \rho = 1 - \exp\left(\frac{(b - c)/(1 - w) + c - b}{2}\right) \quad (\text{S32})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{b + c - (b - c)/(1 - w)}{2}\right). \quad (\text{S33})$$

Therefore, defection will exclude cooperation when

$$\rho < \frac{\kappa_2}{\kappa_1} \quad (\text{S34})$$

$$\iff \exp\left(\frac{(b - c)/(1 - w) + c - b}{2}\right) < \exp\left(\frac{b + c - (b - c)/(1 - w)}{2}\right) \quad (\text{S35})$$

$$\iff bw < c \quad (\text{S36})$$

However, note that the following equality holds whenever $c > 0$:

$$\frac{1}{\rho} < \frac{\kappa_2}{\kappa_1}. \quad (\text{S37})$$

Therefore, when $w > c/b$, we will have

$$\frac{1}{\rho} < \frac{\kappa_2}{\kappa_1} < \rho, \quad (\text{S38})$$

which implies priority effect (equivalently, bistability). Even as $w \rightarrow 1^-$, this inequality holds and therefore, cooperation cannot competitively exclude defection.

Indirect reciprocity. Indirect reciprocity can be described by the following payoff matrix:

$$\begin{pmatrix} b - c & -c(1 - q) \\ b(1 - q) & 0 \end{pmatrix}, \quad (\text{S39})$$

where q represents social acquaintanceship. In this case, the niche and fitness differences correspond to:

$$1 - \rho = 1 - \exp\left(\frac{(b - c)q}{2}\right), \quad (\text{S40})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{2c - (b + c)q}{2}\right). \quad (\text{S41})$$

In this case, the condition for the dominance of defection (and therefore competitive exclusion of cooperation) can be written as:

$$\rho < \frac{\kappa_2}{\kappa_1} \quad (\text{S42})$$

$$\iff bq < c \quad (\text{S43})$$

Moreover, the following inequality always holds as long as $c > 0$.

$$\frac{1}{\rho} < \frac{\kappa_2}{\kappa_1}, \quad (\text{S44})$$

which implies that coexistence between cooperation and defection is not possible. Therefore, when $q < c/b$, then defection will competitively exclude cooperation. Otherwise, the system will enter the priority-effect regime.

Interestingly, when $q \rightarrow 1$, we have

$$\rho = \exp\left(\frac{b-c}{2}\right) \quad (\text{S45})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{c-b}{2}\right) \quad (\text{S46})$$

$$\frac{1}{\rho} = \exp\left(\frac{c-b}{2}\right) \quad (\text{S47})$$

$$(\text{S48})$$

In other words, the following inequality will hold:

$$\frac{1}{\rho} = \frac{\kappa_2}{\kappa_1} < \rho \quad (\text{S49})$$

Network reciprocity. Network reciprocity can be described by the following payoff matrix:

$$\begin{pmatrix} b-c & H-c \\ b-H & 0 \end{pmatrix}, \quad (\text{S50})$$

where $H = [(b-c)k - 2c]/[(k+1)(k-2)]$ and k represents the number of neighbors. In this case, the niche and fitness differences correspond to:

$$1 - \rho = 0, \quad (\text{S51})$$

$$\frac{\kappa_2}{\kappa_1} = \exp(-H + c), \quad (\text{S52})$$

which implies that there is complete niche overlap and that fitness difference decreases with H . Since H is a monotonically decreasing function of k for $k > 2$, fitness difference will increase with k . In particular, cooperation will competitively exclude defection when $H > c$, meaning that:

$$b/c > k. \quad (\text{S53})$$

Group selection. Group selection can be described by the following payoff matrix:

$$\begin{pmatrix} (b-c)(m+n) & (b-c)m - cn \\ bn & 0 \end{pmatrix}, \quad (\text{S54})$$

where m represents the number of groups and n represents the group size. In this case, the niche and fitness differences correspond to:

$$1 - \rho = 0, \quad (\text{S55})$$

$$\frac{\kappa_2}{\kappa_1} = \exp((c - b)m + cn), \quad (\text{S56})$$

which implies that there is complete niche overlap. The fitness difference will increase with n but decrease with m , provided that $b > c$. In particular, cooperation will competitively exclude defection under the following condition:

$$1 + \frac{n}{m} < \frac{b}{c}. \quad (\text{S57})$$

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